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Research Article

Will You Trust Me? Evidence from the Trust Game Between Groups of Friends and Strangers

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Abstract

We report the experimental results from a trust game that was played between friends and strangers. We invited subjects to the laboratory in groups of four friends, who were then assigned across different groups. Some groups of friends stayed together; others were mixed with strangers within their own group or in the opponent group; and some were in pairs of groups where everyone was a stranger to each other. Our results confirm the finding from the previous studies that a pair of friends is more willing to reciprocate a transfer from their friends than from a pair of strangers. We also show that trust and trustworthiness are undermined if the decision is made by a mixed group when compared to a group of friends who only have one friend in the other group. In general, our results suggest that the existence of between-group social ties is insufficient to build trust and trustworthiness. By revealing to a player that their co-player has a friend in the opponent, their group significantly increases trust but not trustworthiness.

JEL Codes: C92, D90

Keywords

Trust game, group decision-making, friends, strangers, social ties

Introduction

Trust is considered to be a fundamental factor in collective decision-making and solving economic issues (Hardin, 2002; Putnam, 2001). It supports mutually

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beneficial relationships and interactions that occur between family members and co-workers (Granovetter, 1983; Mondak & Mutz, 2001), participants of economic exchanges (Fehr et al., 1997; North, 1990), policymakers and taxpayers (Scholz & Lubell, 1998a, 1998b), and communities involved in common pool resource management (Ostrom, 1990). Increased trust leads to more efficient economic outcomes (Zak & Knack, 2001) by reducing risk (Das & Teng, 2001; McLain & Hackman, 1999), information asymmetry and transaction costs (Fukuyama, 1996). In addition, trust increases with a decrease in the social distance and evolution of social networks (Binzel & Fehr, 2013; Etang et al., 2011; Foddy et al., 2009; Johansson-Stenman et al., 2009; Glaeser et al., 2000; Güth et al., 2008; Tanis & Postmes, 2005). However, social connections can be also a source of inefficiency in nonmarket institutions by reducing trade and mobility or by limiting capital flow to a particular social network (Munshi & Banerjee, 2006).

Understanding how to increase trust between individuals is an important question. Previous studies have shown that playing a trust game between a pair of friends increases trust and trustworthiness compared to a game played between strangers (Binzel & Fehr, 2009; Glaeser et al., 2000). Another challenge is how to build trust between groups. Will increased trust and trustworthiness persist between groups of friends when compared to groups of strangers? If we have friends in a different group, then will it help to increase the trust of strangers with whom we make a decision? To address this gap, we report the results from a trust game between groups of two members each. We invited subjects to the laboratory in groups of four friends, who were then allocated across different groups. Some groups of friends stayed together; others were mixed with strangers within their own group or in the opponent group; and some were in pairs of groups where everyone was a stranger to each other. The participants communicated what part of their initial endowment to send to responders within a group using computer chat. We show that friends are more willing to trust and reciprocate to their friends than to a pair of strangers. We also find that revealing to a player that their co-player has friends in the opponent group significantly increases trust but not trustworthiness. Our results show that relative return mostly depends on the decisions made by trustors.

This article makes two main contributions: (a) we examine the effect of stronger social ties/lower social distance with the opponent group on transfer decisions, and (b) we examine the effect of stronger social ties in our own group on transfer decisions. Most studies have examined the impact of real-world social networks on the behaviour of individual decision-makers. For example, it has been shown that pro-social behaviour rises when social distance decreases in a dictator game (Branas-Garza et al., 2010; d'Exelle & Riedl, 2010; Goeree et al., 2010; Leider et al., 2009; Ligon & Schechter, 2012). Furthermore, friends in the role of responders in a power-to-take game playing against a stranger are found to be more emotional in decision-making, punish proposers more than strangers and are more likely to coordinate their punishment without communication (Reuben & van Winden, 2008). Evidence from a microfinance game on loan repayment shows that groups with stronger social ties have higher but less stable group loan repayment when compared to groups without social ties (Abbink et al., 2002).

Our design mostly relates to studies that investigate the relationship between social connections and the behaviour of individual decision-makers in a trust game (Binzel & Fehr, 2009, 2013; Glaeser et al., 2000). In particular, Glaeser et al. (2000) find that the level of trust and trustworthiness increases when a pair of friends plays against each other. Binzel and Fehr (2009) in a binary trust game present similar results (i.e., they show that trust increases with the reduction of social distance). In a later study, the authors introduce a within-subject design to measure trust where a friend is once paired with a stranger and once with a socially close individual (Binzel & Fehr, 2013). Their results show that trust depends on expectations regarding the relative return from a trustee. In particular, a trustor underestimates their friend's trustworthiness, which results in ineffective outcomes. However, no study has examined the impact of the combination of within- and between-group social ties on the behaviour of group members.

We propose additional treatments with a group in which a friend who has at least one friend on the opposite team is paired with a stranger. We refer to this as a 'heterogeneous group'. To the best of our knowledge, no experimental studies have investigated the decisions made by a heterogeneous group. In this article, we address this gap and separately compare the influence of within- and between-group social ties on decisions made in a trust game. In general, whether the choice is made by a pair of friends or a heterogeneous group, we find no differences in trust or relative return when there are two friends on the opposite team. However, trust and trustworthiness decrease when the decision is made by a heterogeneous group when compared to a group of friends who have only one friend in the other group. Our findings indicate that one cannot assume higher levels of trust and trustworthiness based on the existence of social connections between groups.

We find that revealing to the stranger the fact that a player has friends in another group significantly increases trust but not trustworthiness. Our results show that relative return mostly depends on the decisions made by the trustors. In this context, studying the impact of within- and between-group social ties on decisions could help us to understand the conditions under which groups of friends and strangers can trust each other and reach the most effective solutions and exchanges. For example, these issues are experienced by firms that are divided into smaller units as departments between which members have more or less social connections.

Our article also relates to studies that analyse the behaviour of groups in a trust game (Cox, 2002; Kugler et al., 2012; Song, 2008, 2009). However, most studies have focused on the differences between the behaviour of individuals and groups. For instance, Cox (2002) compares the trust and trustworthiness of individuals and groups. The author reports that there are no differences in the amount sent between groups and individuals but groups returned significantly less compared to individuals. In his study, each participant made their decisions in the trust game as an individual and in a group of three through face-to-face discussion. Meanwhile, in Kugler et al.'s (2012) study, each participant played the trust game only once, either as an individual decision-maker or as a group member. In addition, the authors compare asymmetric settings where individuals interact with groups of three. Their results show that groups are less trusting than individuals

but just as trustworthy. Song (2008) shows that participants who make decisions as group representatives are less trusty and trustworthy as opposed to when they are making the analogous decision solely on their own behalf. In her second study, Song (2009) found that groups of three who made consensus decisions were more trusting but less reciprocating when compared to individual decision-makers. Our article differs from previous studies in two ways. We first compare decisions made by groups of friends and strangers in a trust game, and we then include choices made by a heterogeneous group in the analysis (i.e., a stranger paired with a friend who has social connections with the opposite team).

The rest of this article is organised as follows. The second section includes the design and procedures of the conducted experiment. We present our hypotheses in the third section. All of the results are presented in the fourth section, which is followed by the fifth section which contains our conclusions.

Experimental Design and Procedures

The experiment was conducted at the Laboratory of Experimental Economics at the University of Warsaw. A total of 236 subjects participated in 16 sessions. The computerised experimental sessions were run using *z*-tree (Fischbacher, 2007). The participants were 122 females and 114 males, with an average age of 24.77 (±9.02). At the end of the session, the subjects were paid privately in cash. On average, participants earned PLN 34.00 (±9.53), which included a show-up fee of PLN 10 (approximately €2.27). In total, this corresponds to an average payment of €7.72 (±2.17) per participant, based on an exchange rate of PLN 4.4 = €1. Each session lasted approximately 70 min.

The design was based on a trust game (also called an investment game) between groups. In a typical trust game between individuals, there are two players: one player is the sender and the other player is the responder. At the beginning of the experiment, the sender receives initial endowment X, $X > 0$, while the responder receives no money. In the first stage, the sender transfers any amount x that satisfies the condition: $0 \leq x \leq X$. The responder receives triple the amount sent ($3x$) and in the second stage decides how much money to return (y) to the first player, with $0 \leq y \leq 3x$. The sender's final payoff equals the initial endowment reduced by the transferred money to the responder and augmented by the amount received from the responder: $X - x + y$. The responder's final payoff equals the tripled amount sent by the first player minus money returned to them: $3x - y$. The amount x is taken as an indicator of the sender's trust, while the relative return $y/3x$ measures the responder's trustworthiness.

In our study, we conducted a one-shot trust game between groups. There were four players: two served in the role of the sender (A group) and two in the role of the responder (B group). At the beginning of the experiment, A group received PLN 20, while B group received no money. A group had 15 min to make a unanimous decision on what part of PLN 20 (x) to send to the B group ($0 \leq x \leq 20$). The amount sent was tripled and transferred to B group ($3x$). Then, B group made a

Table 1. Summary of Treatments.

Treatment	Description of Treatments Group A—Group B	Total Participants	Total Observations
FF	Friends—friends	32	8
FH	Friends—heterogeneous	36	9
HF	Heterogeneous—friends	36	9
FS	Friends—strangers	32	8
SF	Strangers—friends	32	8
HH	Heterogeneous—heterogeneous	32	8
SS	Strangers—strangers	36	9

Note: A heterogeneous group consists of one friend (who has a friend(s) in the other group) and one stranger.

unanimous decision on how much to return to A group ($0 \leq y \leq 3x$). The participants could communicate only by computer chat within a group and there was no communication between groups. If A group did not decide unanimously on how much to send to B group within 15 min, then the game ended. The members of both groups received only a show-up fee. If A group did not make a unanimous decision within 15 min, then members of the B group would have received only a show-up fee. We proposed seven treatments that differed with the combination of friends and strangers in A and B groups.

Table 1 summarises the treatments. In the group indicated as F, there were two friends, in a group called H there was one friend paired with a stranger, and in group S there were two strangers. We proposed the following treatments: in FF treatment, both A and B groups consisted of two friends. In the A group of FH treatment, there was a group of two friends, and members of the B group were one friend and one stranger. In HF treatment, one friend was paired with a stranger in A group, while the B group consisted of two friends. In FS treatment, a group of friends in the role of senders played against a group of two strangers in the role of responders. SF treatment had two strangers in A group and two friends in B group. HH treatment consisted of two groups of one friend and one stranger. In SS treatment, all four players were strangers. Figure 1 presents the structure of groups in different treatments, where blue dots indicate participants, who have social connections in the real life, and white dots denote strangers.

Participants were recruited by the ORSEE platform (Greiner, 2004). They were invited in groups of four friends who knew each other outside the laboratory.[1] The participants were informed that if they did not show up with the required number of friends, then they would not be able to participate in the experiment. Before starting the experiment, each person from the group of friends drew a number that specified which treatment they were assigned. Next, the participants were invited to the room, where they took a seat at a computer that was marked with their drawn number. All of the computers were separated with special partitions to prevent the participants from viewing each other's displays. The participants were asked not to use their mobile phones and not to communicate with each other in any other way than specified in the rules of the experiment. At the

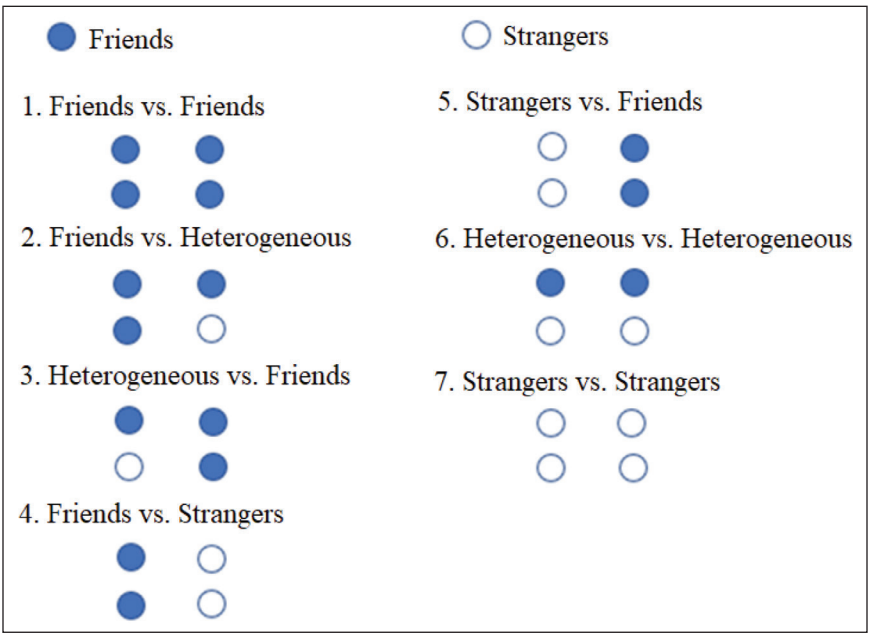

Figure 1. Structure of Groups in Different Treatments.

beginning of each session, the subjects were given written instructions (see Appendix). The participants were informed that they could be matched with their friends in either A or B groups, but this does not need to happen.

The players were informed about whether one or more of their friends were in either group. The participants were asked to enter their name/nickname in the field that appeared on the screen, whereby people from the group of friends could recognise them. In addition, the experimenter informed the participants that their names would not be revealed to anyone outside their group of friends, they would only be revealed if they were co-players.

After entering their name, the participants started the experiment. At the beginning, each participant received information on whether they made a decision either in the A or B group, and whether their friends were in their and/or the opposite group, with an indication of which one exactly.[2] In the first stage of the game, members of the A group had no more than 15 min to chat and make a unanimous decision. At the same time, members of the B group were waiting for A group's decision without any communication. After the transfers from the sender groups were made, the participants were informed about the amount sent by A group and the tripled value. In the next stage, members of the responder group communicated by computer chat to make their decision within 15 min, while A group was waiting. After the decisions were made, all of the participants were informed about the transfers and their final payoffs. Finally, the subjects completed the post-experimental questionnaire. We measured the participants' risk attitudes by a hypothetical lottery question (see Appendix for more details).

Hypotheses

The main focus of our article is to study the influence of within- and between-group social ties on decisions in a trust game.

We formulate the following hypotheses:

H_1: Between-group ties are expected to increase transfers.

We expect that A group of two friends (F) or a friend and a stranger (H) will donate more when more of their friends are in group B. Therefore, we expect that transfers from A group will be larger (Hypothesis 1.1):

FF > FH > FS and
HF > HH > SS[3]

Similarly, we expect the same for transfers from B group, where we hypothesise that transfers will be larger (Hypothesis 1.2):

FF > HF > SF and
FH > HH > SS

We measure between-group ties by the number of connections between groups (i.e., the number of friends in B group).

H_2: Within-group ties will increase transfers if there are between-group social connections. Without social connections with the out-group, transfers will decrease.

In turn, we measure within-group ties by the existence of social connections within a group. Subsequently, we expect that with between-group ties, an A group of friends will make larger transfers when compared to a heterogeneous group, thus (Hypothesis 2.1):

FF > HF
FH > HH

While in group B (Hypothesis 2.2):

FF > FH and
HF > HH

Meanwhile, we believe that without any between-group social connections, a group of friends will send lower transfers to the S group compared to a group of strangers (Hypothesis 2.3):

In group A: FS < SS
In group B: SF < SS

We expect that due to the in-group favouritism (Bernhard et al., 2006; Rusch, 2014) between friends in FS and SF treatments, the F group will be more self-oriented and focused on their own profit maximisation. The concept of in-group favouritism states that people are more interested in the well-being of in-group members when compared to out-group members with whom they do not identify.

Results

Table 2 summarises the mean statistics of transfers between groups for each treatment separately. First, the results show that the transfers from A group to B group are the highest in FH treatment and are equal to 17.78 (±4.41), while the lowest transfers equal to 11.50 (±7.31) and 11.88 (±6.33) are in FS and HH treatments, respectively. Regardless of treatment, the fraction of the amount sent to the B group is between 57% and 89%. Second, in the case of transfers from B group to A group, we observe two distinct levels of trustworthiness. In particular, FF, FH and HF treatments are characterised by the highest level of between- and within-group social connections and indicate a higher level of relative returns. In these groups, trustworthiness equals 0.50, 0.48 and 0.48, respectively. Meanwhile, in the FS, SF, HH and SS treatments, relative return is at a lower level equal to 0.35, 0.35, 0.32 and 0.32, respectively. To assess whether the differences in mean statistics between treatments are statistically significant, we report the results of the Mann–Whitney test (Tables 3 and 5).

Between-group Social Ties

Result 1.1. Between-group social ties increase transfers, but only by groups of friends and not mixed groups. In turn, a group of friends as senders without

Table 2. Mean Statistics by Treatment.

	FF	FH	HF	FS	SF	HH	SS
Average amount sent by group A (x)	15.63 (6.23)	17.78 (4.41)	14.78 (8.24)	11.50 (7.31)	15.63 (5.63)	11.88 (6.33)	14.00 (7.33)
Median x	20	20	20	12	17.5	11	18
Fraction of amount sent $x/20$	0.78	0.89	0.74	0.58	0.78	0.59	0.70
No. of groups with $x = 20$	5	7	6	2	4	2	4
Average return (y)	23.25 (9.72)	25.52 (8.89)	21.10 (13.65)	12.09 (12.80)	16.20 (13.55)	11.50 (12.55)	15.13 (12.21)
Average $y/3x$	0.50	0.48	0.48	0.35	0.35	0.32	0.32
No. of observations	8	9	9	8	8	8	9

Note: The values in the parentheses indicate the standard deviation.

Table 3. *Z*-scores from the Mann–Whitney *U* Test: Comparing Mean Transfer x (from A to B group), by Treatments.

	FH	HF	FS	SF	HH	SS
FF	–0.78	0.11	1.32	0.18	1.11	0.58
FH		0.73	2.09**	1.02	1.99**	1.36
HF			1.12	0.16	1.02	0.49
FS				–1.24	–0.11	–0.84
SF					1.19	0.41
HH						–0.64

Note: **Indicates variables significant at the 5% level.

Table 4. The Results from OLS Regression with the Dependent Variable Equal to Transfer x from A Group.

	Model (1)	Model (2)	Model (3)
Transfer A	All Treatments	FF, FH, FS Treatments	HF, HH, SS Treatments
No. of characters	0.001 (0.001)	–0.006** (0.003)	0.004 (0.004)
Difficulties in decision-making	–0.99 (0.66)	–1.43* (0.81)	–1.12 (1.16)
Treatment FF	–	–	
FH	0.38 (2.13)	0.27 (1.71)	
HF	–2.55 (2.18)		–
FS	–5.49** (2.27)	–4.67** (1.91)	
SF	–1.62 (2.21)		
HH	–4.80** (2.17)		–2.53 (2.80)
SS	–3.86* (2.17)		–1.56 (2.47)
Female	–1.69 (1.21)	–3.42* (1.72)	–1.61 (2.10)
Age	–0.27*** (0.07)	–0.34*** (0.07)	–0.18 (0.16)
Risk measure	0.04 (0.43)	0.45 (0.57)	–0.26 (0.83)
Cons.	25.98*** (3.19)	29.65*** (3.79)	21.63*** (5.56)
No. of observations	118	50	52
Prob. > *F*	0.0004	0.0000	0.6346
R^2	0.2638	0.5780	0.1061

Note: Standard errors in parentheses. ***Indicates variables significant at the 1% level, **at the 5% level, and *at the 10% level.

Table 5. *Z* Scores from the Mann–Whitney *U* Test: Comparing Relative Return $\left(\frac{y}{3x}\right)$, by Treatments.

	FH	HF	FS	SF	HH	SS
FF	0.79	0.97	2.15**	1.28	2.28**	1.96**
FH		0.26	2.16**	0.92	2.25**	1.88*
HF			1.57	0.55	1.75*	1.41
FS				–0.23	0.41	–0.23
SF					0.80	0.11
HH						–0.75

Note: **Indicates variables significant at the 5% level and *at the 10% level.

between-group social ties is less trusting towards responders when compared to a group of friends that has at least one friend in the opposite group.

Average transfers made by groups of friends in FF, FH and FS treatments are equal to 15.63 (±6.23), 17.78 (±4.41) and 11.50 (±7.31), respectively (Table 2). To verify that between-group social ties affect trusting behaviour, we compare transfers from sender groups in FF, FH and FS treatments. The Mann–Whitney test indicates statistically higher transfers made by sender groups in FH treatment compared to FS treatment (Table 3). In addition, 78% of A groups in the FH treatment decided to send the whole amount to the responder group compared to 25% of A groups in the FS treatment (Fisher's exact $p = .04$), resulting in a higher average transfer from A group in the former than in the latter treatment.

In Table 4, we report the results from OLS regressions with transfer from A group as a dependent variable. As predictor variables, we included a treatment variable, a variable describing difficulties in decision-making and the number of characters in the chat. In addition, we control for the effects of gender, age and risk preferences. Model 2 presents the results for the subsample of FF, FH and FS treatments, while Model 3 is based on a subsample of HF, HH and SS treatments. The results in Table 4 (Model 2) show that the coefficient corresponding to the FS treatment variable is negative and significant, which is evidence that the expected transfer A in the FS treatment is lower by 11.5 when compared to the FF treatment. We do not observe statistically significant differences in the transfers from the A group between FF and FH treatments.

The average transfer A in HF, HH and SS treatments are equal to 14.78 (±8.24), 11.88 (±6.33) and 14.00 (±7.33), respectively. The differences between these amounts are statistically insignificant according to the Mann–Whitney test (Table 3). The results from OLS regression in Table 4 (Model 4) also show that coefficients corresponding to treatment variables are statistically insignificant. We find that 67% of A groups in HF treatment send the maximum amount to the responder group when compared to 25% of the corresponding group in HH treatment, which is statistically different according to Fisher's exact test ($p = .086$).

Result 1.2: Responders' trustworthiness in FF treatment is significantly higher than in HF and SF treatments.

The results in Table 2 show that relative returns are equal to 50%, 48% and 35% in FF, HF and SF treatments, respectively. Table 5 reports the results from Mann–Whitney tests, which show that there is no statistically significant difference in trustworthiness between these treatments.

In Table 6, we present the results from OLS regressions with the dependent variable equal to the relative return. As predictor variables, we included transfer from the A group, a variable describing difficulties in decision-making, the number of characters in the chat and a treatment variable. In addition, we control the effect of gender, age and risk preferences. In Model 5, we present the results from OLS regression run on the sample of data from FF, HF and SF treatments, while Model 6 is based on a subsample of FH, HH and SS treatments. Evidence from

Table 6. The Results from OLS Regression with the Dependent Variable Equal to the Relative Return.

	Model (4)	Model (5)	Model (6)
Trustworthiness	All Treatments	FF, HF, SF Treatments	FH, HH, SS Treatments
Transfer from A group	0.02*** (0.00)	0.01** (0.01)	0.02*** (0.00)
No. of characters	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Difficulties in decision-making	0.01 (0.02)	0.02 (0.02)	–0.01 (0.03)
Treatment FF	–	–	
FH	–0.07 (0.05)		–
HF	–0.12** (0.06)	–0.14* (0.07)	
FS	–0.18*** (0.06)		
SF	–0.24*** (0.06)	–0.26*** (0.07)	
HH	–0.20*** (0.06)		–0.09* (0.05)
SS	–0.13** (0.06)		–0.04 (0.05)
Female	–0.02 (0.03)	–0.12* (0.06)	0.04 (0.04)
Age	–0.02 (0.03)	0.00 (0.00)	–0.00** (0.00)
Risk measure	0.00* (0.01)	0.04** (0.02)	0.02 (0.01)
Cons.	0.14 (0.10)	0.18 (0.18)	0.23** (0.011)
No. of observations	112	48	50
Prob. > *F*	0.0000	0.0003	0.0000
R^2	0.4961	0.4999	0.6076

Note: Standard errors in parentheses. ***Indicates variables significant at the 1% level, **at the 5% level and *at the 10% level.

Model 5 indicates that the predicted level of trustworthiness is significantly lower in the HF treatment and SF treatment when compared to the FF treatment.

We also compared differences between these treatments in frequencies when trustworthiness is lower than 0.5 and equal to or higher than 0.5. Fisher's exact tests showed that the relative return equal to or higher than 0.5 is more frequent in FF ($p = .05$) and HF treatments ($p = .03$) when compared to the SF treatment.

Result 1.3. Relative return is higher in the FH treatment compared to the HH and SS treatments, while there are no differences in trustworthiness between the HH and SS treatments.

According to the results from Table 2, relative returns in FH, HH and SS treatments are equal to 0.48, 0.32 and 0.32, respectively. We find that trustworthiness in FH treatment is significantly higher when compared to HH ($p < .05$) and SS ($p < .1$) treatments (Table 5). This finding is also supported by the evidence in Table 6 (Model 6). In particular, the variable corresponding to treatment effects is significant and negative, which indicates that the predicted level of trustworthiness is lower in the HH treatment than in the FH treatment. In addition, Fisher's exact tests support these results, showing that a relative return equal to or higher than 0.5 is more frequent in the FH treatment when compared to the HH ($p = .03$) and SS treatments ($p = .06$).

Within-group Social Ties

Result 2.1. Within-group social ties increase the level of trust if there is one friend in the responder group.

Table 2 shows that the mean transfer from A group in the FH treatment equals 17.78 (±4.41) when compared to 11.88 (±6.33) in the HH treatment. According to the results from Table 3, this difference is statistically significant ($p < .05$). This result is also supported by Fisher's exact test, which showed that transfers from A group equal to the maximum amount are more frequent in the FH compared to the HH treatment ($p = .03$).

We find that transfers made by a group of senders in the FF treatment do not differ from transfers in the HF treatment. In particular, the average amount sent by group A in the FF treatment equals 15.63 (±6.23), while in the HF treatment, it is equal to 14.78 (±8.24). The differences in mean transfers made by sender groups in FS and SS treatments are not statistically significant. Mean transfer A in the FS treatment equals 11.50 (±7.31), whereas in the SS treatment, it equals 14.00 (±7.33).

Result 2.2. Within-group social ties increase trustworthiness if there is one friend in the sender group.

The results show that the trustworthiness between the HF and HH treatments is equal to 48% and 32%, respectively (Table 2). This difference is statistically significant ($p < .1$). In particular, Fisher's exact test confirmed that trustworthiness equal to or higher than 0.5 is more frequent in the HF treatment compared to the HH treatment ($p = .057$). We do not find evidence that social ties within groups affect trustworthiness if there is a group of friends as senders. In particular, the results in Table 2 show that the relative return is equal to 50% in the FF and 48% in the FH treatments. In addition, we do not find differences in trustworthiness between SF and SS treatments. The relative returns in these treatments are equal to 35% and 32%, respectively.

Chat Analysis

In this subsection, we describe the most important differences that appeared in the discussion between group members and which could have potentially influenced the players' decisions. Members of Group A needed an average of 5.64 (3.66) min to make a decision, while members of Group B needed 4.83 (3.18) min. This difference and the difference in the division into treatments were not statistically significant. Table A1 in Appendix presents the categories of coding messages between players and Table A2 illustrates the frequencies of coding '1' for each category and treatment. Table A3 summarises the results from the Tobit regressions. In particular, the dependent variable in Models A1 and A2 equals the transfer from A group, while in Models A3 and A4, it is trustworthiness.

We include categories of coded records from the chat as explanatory variables (Models A1 and A3). Models A2 and A4 illustrate the results from the regressions run on the sample of the HF and HH treatments (Model A2), and the FH and HH treatments (Model A4).

Evidence from Model A1 indicates that transfer A increases when a player proposes to send the maximum amount. In favour of this, the coefficient related to the variable C1 is positive and statistically significant. This result may explain higher transfers from A groups in FF and FH treatment when compared to the FS treatment, considering 75% and 89% of A groups in former treatments in which such a proposition appeared relative to 25% of corresponding groups in the FS treatment.

In heterogeneous groups, one player had at least one friend on the opposing team. In Table A3, the coefficient corresponding to revealing this information to a stranger is significant and positive (Model A2). This indicates that providing this information increases trust but has no effect on trustworthiness (Model A4). We find that a friend revealed to the stranger that has another friend(s) in the responder group in 67% of the A groups in the HF treatment compared to 38% of the groups in the HH treatment. This difference is statistically insignificant. In contrast, we observe that a friend decided to reveal that they have a friend(s) in the opposite team in only 22%–25% of heterogeneous B groups. In this case, we find that providing this information does not influence the level of trustworthiness.

Our results show that transfers from A group are higher in the FH treatment when compared to the HH treatment. This outcome is supported by the fact that a player offered to send the maximum amount in 89% of A groups in the FH treatment when compared to 25% of A groups in the HH treatment (Fisher's exact test $p = .02$). Furthermore, we find that a friend offered to send a lower amount than a stranger to the opposite group in 50% of heterogeneous A groups in the HH treatment, despite having a friend in B group.

Detailed analysis of chat records showed that friends considered not returning any amount in 56% of B groups in the HF treatment to save as much money as possible and share this amount with their friend from A group after the experiment, which to some extent indicates an in-group favouritism. However, none of these groups decided to follow this rule.

Conclusions

In this article, we experimentally investigated the impact of between- and within-group social ties on trust and trustworthiness. Our study provides an additional piece of evidence and possible explanations regarding the decisions made by friends and strangers. In general, it has been documented that trust increases with lower social distance (Binzel & Fehr, 2013; Etang et al., 2011; Johansson-Stenman et al., 2009; Leider et al., 2009). In particular, studies show that trust and trustworthiness increase between a pair of friends when compared to a pair of strangers (Binzel & Fehr, 2009, 2013; Glaeser et al., 2000). We provide additional

evidence on the behaviour of groups of friends and strangers in a trust game. We show that a group of friends is more trusting if there are between-group social connections, while a group of friends is less trusting of a pair of strangers. These results suggest that a combination of within- and between-group ties increases expectations for higher returns and supports the previous results by Binzel and Fehr (2013) for individual decision-makers. In addition, we find that a relative return from a group of friends is higher only if there are two friends in the sender team compared to a situation when there is one friend or a pair of strangers in the sender team.

We also introduce a heterogeneous group that consists of a friend who has at least one friend on the opposite team and is paired with a stranger. To date, no study has examined the effects of a mixed group on trust and trustworthiness. Our results show that revealing the fact that a player has friends in another group to the stranger significantly increases trust but not trustworthiness. Furthermore, we show that between-group social connections do not influence trust if a friend is paired with a stranger in the sender group. However, the trustworthiness of a heterogeneous group is higher when there are two friends in the sender group when compared to two strangers or a mixed group as senders. The possible explanation for this outcome is that the relative return mostly depends on transfers from sender groups, which is supported by the fact that there were no differences in payoffs between A- and B groups in any of the treatments. In general, our results suggest that the existence of between-group social ties is insufficient to assume higher trust and trustworthiness.

In the context of within-group social ties, we find that there are no differences in trust and relative return when there are two friends on the opposite side, regardless of whether the decision is made by a pair of friends or a pair of a friend and a stranger. However, trust and trustworthiness decline when the decision is made by a heterogeneous group when compared to a group of friends who have only one friend in the opposing group. This outcome also confirms that having social connections with the opposite group may not necessarily lead to increased trust and trustworthiness.

We also hypothesised that without between-group social connections, a group of friends will transfer and reciprocate less when compared to a group of strangers. We expected that this result would be followed by a pattern called in-group favouritism, which states that people are more pro-social towards in-group members when compared to members of other groups (Bernhard et al., 2006; Rusch, 2014). In this case, a group of friends could have been more self-oriented than a group of strangers. However, our results do not support this effect. In particular, we do not observe differences in trust and relative return between a pair of friends and a pair of strangers who play against two strangers.

Our results indicate that the average fraction of initial endowment sent by a group of trustors varies between 0.58 and 0.89. A meta-analysis of trust games between individuals conducted in different countries has shown that this average fraction equals 0.5 (Johnson & Mislin, 2011). We might expect that our outcomes would be lower because groups are considered to behave more in line with game theoretic equilibrium and thus are more rational and selfish (Kugler et al., 2012).

Furthermore, the results presented by Kugler et al. (2007) indicate that a group of three sends to the other group on average 44% of their initial endowment in a between-group trust game. In our study, high transfers from A groups might be explained by an effect of 'indirect reciprocity', in the sense that participants came to the experiment in groups of four friends and were divided into specific treatments. In this case, the decisions of those participants who did not play with their friends might have been influenced by affective priming regarding their feeling of being an in-group member by coming and participating in the session with their friends.

Examining the impact of within- and between-group social connections on the behaviour of agents can provide useful explanations of the conditions under which the groups are more willing to trust, reciprocate and cooperate, which consequently contributes to the improvement of social welfare. For instance, it has been argued that social networks play an important role when formal contract enforcements are absent by increasing the cooperation between partners (Chandrasekhar et al., 2018). This result confirms other theories that highlight the value of social networks in fostering mutually beneficial exchanges (Jackson et al., 2012). Our results provide additional observations in this matter. In particular, groups that consist of friends and strangers may not necessarily reach a sufficient level of cooperation with other groups. Therefore, future studies should focus on investigating the effects of pairing friends and strangers in different economic settings, such as a dictator game or a public goods game.

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Notes

1. In the post-experimental questionnaire, we asked how many months they had known each other and how often they met during a year to determine the strength of their relationships. The participants answered these questions separately to each friend with whom they were playing with.
2. For example, a friend from A-group in the FF treatment received an information: 'You are in group A. In group A there is your friend called name_1. In group B, there are your friends called name_2 and name_3'. A friend from A-group in the HH treatment

received an information: 'You are in group A. In group A there are none of your friends with whom you came for the experiment today. In group B, there is your friend called name_1'. A stranger (a subject who does not have her friends among the co-players) was informed that: 'In group A and B there are none of your friends with whom you came for the experiment today'.

3. We can treat SS treatment as HS treatment because in this case we assume that in heterogeneous A group, there is one friend paired with a stranger; however, this friend does not have any social connections with the outgroup. Therefore, it brings us to SS treatment.

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Appendix

Table A1. Coding Table.

	Description
Code	Group A
C1-A	Proposition to send whole amount
C2-A	Proposition to send nothing
C3-A	Hope for equal split
C4-A	Hope for return more than 0
C5-A	Agreement was reached on a first proposal
C6-A	First proposition was equal to 10
C7-A	Considering different scenarios
C8-A	Appeal to fairness
C9-A	Using goodwill as an argument to send whole amount with expectation of reciprocity
C11-A	Thinks that other group will not return much/do not trust other group
C13-A	More egoistic person dominates more altruistic
C14-A	More altruistic person dominates more egoistic
C15-A	Reveals that has friend(s) in the other group
	Group B
C2-B	Proposition to send nothing
C3-B	Proposition of equal split
C5-B	Agreement was reached on a first proposal
C7-B	Considering different scenarios
C8-B	Appeal to fairness
C10-B	Positive comment about group A (e.g., appreciates well behavior of A)

(Table A1 continued)

(Table A1 continued)

	Description
Code	Group A
C12-B	Negative comment about group A (e.g., regrets the decision made by A)
C13-B	More egoistic person dominates more altruistic
C14-B	More altruistic person dominates more egoistic
C15-B	Reveals that has friend(s) in the other group

Table A2. Frequency of Coding '1'.

	FF	FH	HF	FS	SF	HH	SS
				Group A			
C1-A	0.75	0.89	0.67	0.25	0.88	0.25	0.56
C2-A	0.00	0.00	0.22	0.25	0.00	0.00	0.44
C3-A	0.50	0.56	0.44	0.63	0.25	0.13	0.44
C4-A	0.63	0.44	0.33	0.38	0.38	0.25	0.39
C5-A	0.63	0.33	0.22	0.00	0.38	0.38	0.33
C6-A	0.25	0.00	0.11	0.00	0.25	0.25	0.11
C7-A	0.13	0.33	0.22	0.38	0.00	0.00	0.56
C8-A	0.25	0.11	0.11	0.00	0.00	0.00	0.00
C9-A	0.00	0.22	0.00	0.13	0.13	0.00	0.33
C11-A	0.25	0.11	0.00	0.25	0.13	0.25	0.11
C13-A	0.13	0.00	0.11	0.00	0.13	0.13	0.11
C14-A	0.00	0.00	0.11	0.00	0.13	0.38	0.11
C15-A	–	–	0.67	–	–	0.38	–
				Group B			
C2-B	0.25	0.22	0.56	0.25	0.50	0.50	0.22
C3-B	0.50	0.89	0.44	0.38	0.50	0.38	0.67
C5-B	0.50	0.22	0.33	0.50	0.25	0.25	0.56
C7-B	0.00	0.22	0.22	0.00	0.00	0.00	0.11
C8-B	0.00	0.33	0.22	0.13	0.38	0.13	0.22
C10-B	0.25	0.33	0.22	0.13	0.38	0.13	0.11
C12-B	0.13	0.11	0.11	0.13	0.25	0.13	0.11
C13-B	0.13	0.00	0.00	0.00	0.25	0.13	0.11
C14-B	0.00	0.00	0.00	0.00	0.25	0.00	0.22
C15-B	–	0.22	–	–	–	0.25	–

Note: A/B indicator in the first column refers to the results from group A or group B, respectively.

Table A3. Results from Tobit Regression with the Dependent Variable Equal to Transfer from A Group (Models A1 and A2) and Trustworthiness (Models A3 and A4).

	Model (A1)	Model (A2)	Model (A3)	Model (A4)
	A Group Decisions		B Group Decisions	
	All Treatments	HF, HH Treatments	All Treatments	FH, HH Treatments
C1	13.74*** (2.13)			
C2	–3.43 (2.67)		–5.17 (5.28)	
C3	–2.44 (2.30)		21.57*** (5.73)	
C4	1.40 (2.58)			
C5	0.81 (2.20)		2.22 (5.31)	
C6	–3.59 (2.25)			
C7	–1.27 (2.31)		14.22 (9.58)	
C8	1.32 (3.37)		10.76 (7.16)	
C9	7.41* (3.98)			
C10			11.04 (6.69)	
C11	–6.06*** (2.27)			
C12			–13.31* (7.14)	
C13	–5.14** (2.49)		–11.82 (9.58)	
C14	1.46 (2.88)		–9.50 (10.85)	
C15		16.22** (6.94)		8.79 (16.15)
Cons.	11.010*** (1.93)	9.47** (4.06)	6.48 (4.99)	19.47** (7.49)
No. of observations	59	17	58	17
Prob. > *F*	0.000	0.0102	0.000	0.5756
Pseudo R^2	0.319	0.0835	0.1931	0.0049

Notes: Standard errors in parentheses. ***Indicates variables significant at the 1% level, **at the 5% level and *at the 10% level.

Post-experiment questionnaire:

1. Are you: male/female
2. Age
3. Which is your major: economics/others/never studied
4. Lottery question (hypothetical): Imagine that you receive PLN 100 from us. You have to decide how much of this amount (from 0 to 100 PLN) you would like to spend on participating in the following lottery: you have 50% chance to lose the allocated amount and you have 50% chance to win three times the amount wagered. How much of PLN 100 would you spend on this lottery?